Homework 4

Solution Key (Revenue-Share HHI, CR2, CR4)

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Definitions

Let firm i have revenue $R_i = p_i \cdot q_i$.
Let total market revenue be VM (column `VM`).

Define **revenue share** as:

$$sh_i = \frac{R_i}{VM} = \frac{p_i q_i}{VM}$$

- **Herfindahl–Hirschman Index (HHI, 0–10,000 scale)**

$$HHI = \sum_{i=1}^{N} (100 \cdot sh_i)^2$$

- **Concentration Ratio (CR_k)**

$$CR_k = \sum_{j=1}^{k} sh_{(j)} \quad \text{where } sh_{(1)} \geq sh_{(2)} \geq \cdots$$

  - **CR2:** sum of top two revenue shares
  - **CR4:** sum of top four revenue shares

  Use revenue shares for **all** calculations:

$$sh_i = \frac{p_i q_i}{V_M}.$$

Mapping to Excel Columns

Per month (row):

Revenues

- `r1 = p1*q1` (Jeminy)

- `r2 = p2*q2` (**Cousins**)
- `r3 = p3*q3`
- `r4 = p4*q4`
- `r5 = p5*q5`
- **All Others:**
  `r_others = MAX(0, VM - (r1 + r2 + r3 + r4 + r5))`

Shares

- `sh_i = r_i / VM` for each of the 5 named firms (+ Others if `r_others > 0`)

Excel tips (replace `shares_range` with the cell range of shares):

- `HHI = SUMPRODUCT((100*shares_range)^2)`
- `CR2 = SUM(LARGE(shares_range,{1,2}))`
- `CR4 = SUM(LARGE(shares_range,{1,2,3,4}))`

  Do **not** use `QM` for HHI/CR — we are using **revenue** shares, not quantity shares.

Graphs Over Time

Create a Year–Month column and compute monthly **HHI**, **CR2**, and **CR4** using revenue shares. Produce three line charts: HHI, CR2, and CR4 vs. `date`.

Interpretation Thresholds

DOJ/FTC HHI screens (0–10,000 scale):

- **Unconcentrated:** HHI < **1500**
- **Moderately concentrated: 1500–2500**
- **Highly concentrated:** > **2500**

CR rules of thumb:

- **CR4 < 40%** → fragmented / competitive
- **CR4 40–60%** → loose oligopoly
- **CR4 > 60%** → tight oligopoly
- **High CR2** (> 40–50%) → strong duopoly tendency

Use both **levels** and **trends** to describe market structure.

Answering Management's Questions

- **HHI over time:** Report the level and trend. Compare to thresholds above.
- **CR2/CR4 over time:** Highlight concentration patterns. High/stable CRs indicate persistent market power.
- **Perfect competition?**
  - Only plausible if HHI is low (**<1500**) **and** CR4 < 40%.
  - Otherwise, the market is likely **oligopolistic**, not perfectly competitive.
- **Cousins' position:**
  Report

  $$sh_{\text{Cousins}} = \frac{p_2 q_2}{VM}$$

  each month, note if it is a top-2 or top-4 firm, and whether its share is increasing or decreasing over time.

Common Pitfalls

- Forgetting **Others** when `VM > Σ(p_i q_i)` → underestimates concentration.
- Mixing units (quantities instead of revenues).
- Shares must sum to 1 for accurate results.
- HHI/CR should always be calculated **monthly**, then graphed over time.